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Week 1

Project Plan 12th June.

The main focus will be increasing the spatial resolution of wildfire satellite images. Step 1 of the big project. If we have time we can go to step 2.

Paper to follow:

https://www.mdpi.com/2072-4292/16/4/715

https://www.sciencedirect.com/science/article/abs/pii/S092427162100160X?via%3Dihub
(30m spatial resolution)

1. GOES vs **MTG**. If it's in Google earth engine it's easier to collect data. What bands are we using? Etc
2. Sentinel-2(20m) vs Landsat(30m) what do we want?
3. Cascade approach or direct approach, or both and compare

Coarse Input Data: GOES vs MTG

	GOES	**MTG**
Spatial Res	2km	Visible and NIR: 1km IR: 2km (Active Fire Monitoring EO:EUM:DAT:0682)
Temporal Res	10 mins	Every 10 mins
Bands	3.9μm	FCI channel IR-3.8μm
Spatial coverage	The Americas (North and South)	Lat: -67.5° to 67.5°, Lon: -67.5° to 67.5° (EU, Africa, Atlantic Ocean, parts of Middle East and South America?)

Temporal extent	GOES-16: May 2017 - Now GOES-17: 2018 - now GOES-18: Oct 2022 - Now	Jul 2025 - Now
Access	GEE API	EUMETSAT API
Advantages	Can access via GEE API	Directly relevant to the ESA project. Greater spatial extent
Limitations	Only covers the Americas	Data only available from July 2025
Relevant links	GEE GOES-16 FDCF Data Guide	Data guide

- Since GOES likely will have more wildfire data (or could check if the spatial breadth of the MTG compensates for the short time period), we could initially work with GOES as our coarse input data. If time permits, we can try tuning the model with MTG?
- Alternatively we can combine MTG and MSG data to get a larger dataset for MTG (provided that the relevant data have the same spatial resolution. The IR bands of MSG are 3km while 1/2km for MTG.
- For MTG, we find that some bands have different resolution (0.5, 1, 2km) due to the different modes (High Resolution Fast Imagery (HRFI))
- Fire Radiative Power - MTG have data available in 1km from May 2026

High-res Target Data: Landsat vs Sentinel-2

	Landsat	**Sentinel-2**
Spatial res	30m	20m
Temporal res	8/16 days	5 days
Relevant Bands	Band 6 (1.6 μm), Band 7 (2.2 μm)	Band 11 (1.6 μm), Band 12 (2.2 μm)
Temporal extent	Landat 8: Apr 2013 - present Landsat 9: Sept 2019 - present In NASA FIRMS: June	March 2017 - present

	2022 - present	
Spatial extent	NASA FIRMS: US/Canada only	global
Access	GEE	Copernicus Data Space Ecosystem (API)
Advantages	Contains thermal infrared band Fire dataset is available	Slightly higher resolution
Limitations	NASA FIRMS only have data from 2022, and only available for US/Canada	Lacks thermal infrared bands. As far as I can find, there is no clear 'fire' dataset that I could use

- Active fire detection in Landsat-8 imagery: A large-scale dataset and a deep-learning study Only fire data for Aug/Sept 2020 (worldwide).

Intermediate Resolution Data

- VIIRS:
  - Resolution: 375m
  - Access via NASA FIRMS
  - Temporal Res: Daily
- Main problem: when doing the cascaded approach, we need to spatially and temporally match VIIRS with the high-res target (landsat or Sentinel). There could potentially be a temporal mismatch

Summary: Likely use GOES for input coarse data, VIIRS for intermediate, and Landsat for 'ground truth'

Main Research

The research will focus on diffusion-based models due to their probabilistic nature, which can be more useful for capturing the stochastic nature of wildfire dynamics than deterministic models in risk assessments (Yu et al.).

I have chosen to compare the following architectures:

- **CredDSR**: to deal with the problem of spatial ambiguity of downscaling low to very high resolutions, CredDSR's cross resolution distance aware mechanism calculates the

distance between the low and the high-res cells to guide the downscaling. Terrain difficulty loss is used in the paper to give more learning weights to difficult terrains.
    - Enforces explicit physical distance constraints within the attention mechanism (CreAM) and uses a Terrain Difficulty Perception Loss for complex terrain. This method focuses on anchoring attention weights to true geographic proximity to prevent spatial hallucinations.
- **TAUT-based architecture with MBTS:** wildfires are constrained by physical variables such as topography. TAUT uses a multi-branch terrain adaptive module (MBTA) which extracts multi-scale topographical features such as slope etc.
    - Employs MBTA module to explicitly align dynamic atmospheric data with static geographic features via parallel spatial/channel attention in the paper. This can be applied to our wildfire case. This method focuses on parallel fusion pathways that handle multi-modal covariates.

Alternatively: change the below table so that we don't use DDPM for the CredDSR/TAUT methods, and just use the methods as itself.

Experiment	**Architecture**	**Pipeline Structure**	**Input Resolution**	**Output resolution**	**Primary Research Objective**
E1	Conditional DDPM + simple CNN/ViT UNet	Direct	2km (GOES)	30m (Landsat, Pereira et al.)	The baseline. Tests how a vanilla baseline handles big resolution leaps.
E2	Conditional DDPM + simple CNN/ViT UNet	Cascaded	Stage 1: 2km -> 375m Stage 2: 375m -> 30m	30m (Landsat, Pereira et al.)	Tests if breaking the downscaling into two steps improves the baseline model's accuracy.
E3	Conditional DDPM with TAUT-based UNet transformer backbone	Direct	2km (GOES)	30m (Landsat, Pereira et al.)	
E4	Conditional DDPM with TAUT-based	Cascaded	Stage 1: 2km -> 375m Stage 2:	30m (Landsat, Pereira et al.)	

	UNet transformer backbone		375m -> 30m		
E5	Conditional DDPM with CredDSR UNet backbone	Direct	2km (GOES)	30m (Landsat, Pereira et al.)	
E6	Conditional DDPM with CredDSR Unet backbone	Cascaded	Stage 1: 2km -> 375m Stage 2: 375m -> 30m	30m (Landsat, Pereira et al.)	
E7	Conditional DDPM with … backbone	Direct	2km (GOES)	30m (Landsat, Pereira et al.)	
E8	Conditional DDPM with … backbone	Cascaded	Stage 1: 2km -> 375m Stage 2: 375m -> 30m	30m (Landsat, Pereira et al.)	

So for the Cascade approach, first you train a model that goes from 2km to 375m. Then we use that model to increase the res of Goes so we can match Goes and Landsat.

WEEK 2

Objectives:

1. Only satellites or more data? (Landcover, DEM, …). Please select the bands we are using in the satellites.
2. Evaluation metric (what we use to know if the result is good, MSE? Something that measures the blurriness)
3. Decide 3 architectures to try.
4. Ask gemini for a plan to train the model in Google Cloud Platform. So ask gemini what is the best way to ingest the data (data pipeline big query ? or?). IRP plan

Inputs/outputs/possible covariates:

- **GOES-16/17/18** bands (from https://www.sciencedirect.com/science/article/pii/S2666017221000031):
    - Active fire bands: 3.80–3.99 μm and 10.82–11.60 μm at 2km
    - Alternatively, use Level 2 Fire Mask (FSH) or Fire Radiative Power (FRP) bands directly, at 2 km.
- **VIIRS** (375m):

    - Use NASA FIRMS data, which uses I-bands: Bright_ti4, Bright_ti5, frp (fire radiative power) for thermal anomaly detection.
- **Topography:** Copernicus DEM GLO-30 (30m)
- **Landcover:** ESA WorldCover 10m v100 **(10m) (Here it tells you the water bodies)**
- **NDWI (Normalised Difference Water Index):** Landsat Collection 2 Tier 1 Level 2 32-Day NDWI Composite **(30m) (dont need this)**
- **NDVI (Normalised Difference Vegetation Index):** Landsat Collection 2 Tier 1 Level 2 8-Day NDVI Composite **(30m)**
- **Global inland water: GLCF:** Landsat Global Inland Water **(30m)**
- **Buildings: GHSL:** Global settlement characteristics (10 m) 2018 (P2023A) **(10m)**
- **Landsat** (Pereira et al. dataset) (30m)
    - Uses 10 bands: Channels b1 to b11 except b8: Green, Red, Near-Infrared (NIR), Shortwave Infrared 1 (SWIR 1), Shortwave Infrared 2 (SWIR 2), Cirrus, Thermal Infrared 1 (TIRS 1), Thermal Infrared 2 (TIRS 2)

Metrics

- IOU/F1 (binary mask):
    - IOU Quantifies the overlap between the predicted fire mask and the ground truth mask.
    - F1: The harmonic mean of precision (how many predicted fires were real) and recall (how many real fires were detected).
- (PSNR/SSIM: standard image-based metrics)
- FID: evaluates the quality of generative outputs by comparing the statistical distribution of generated images to real images.
- MSE/RMSE
- CG-FG MSE (Coarse-Grid to Fine-Grid Mean Squared Error) and SR-FG MSE (Super-Resolution to Fine-Grid Mean Squared Error)